INCH-POUND

MIL-C-24643/21C
14 March 1994
SUPERSEDING
MIL-C-24643/21B
3 October 1986

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPES LSTCJU AND LSTCTU
(INCLUDING VARIATION LSTCJA AND LSTCTA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

REQUIREMENTS:

Qualification required.

Construction (watertight with circuit integrity)

First - One iron and one constantan conductor in type LSTCTU, for type J thermocouple in accordance with ANSI MC96.1; one copper and one constantan in type LSTCTU, for type T thermocouple in accordance with ANSI MC96.1; (see table I for size). All conductors uncoated.
Second - Extruded Silicone rubber insulation (see table I for diameter).
* Third - Glass braid, providing 80% minimum coverage.
Fourth - Braid covering. Printed identification as follows: Type LSTCJU shall have the legend "8 (GRAY)" printed on the iron conductor and the legend "3 (RED" printed on the constantan conductor. Type LSTCTU shall have the legend "6 (BLUE)" printed on the copper conductor and the legend "3 (RED)" printed on the constantan conductor (see table I for diameter).
* Fifth - The two conductors shall be cabled together with a lay of 16 to 24 times the pitch diameter. Fillers shall be used as necessary to form a firm, well rounded assembly.
Sixth - A binder or combination binder and barrier.

Construction variant for types LSTCJU and LSTCTU

* Seventh - Cross-linked polyolefin jacket. Nominal thickness 0.050 inch.

Construction variant for types LSTCTA and LSTCJA

* Seventh - Cross-linked polyolefin jacket. Nominal thickness 0.050 inch.
* Eighth - Braided metal armor

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* TABLE I. Details.

Military part no. M24643/21	Type and Size	Number of strands and strand diameter (inch)	Diameter over insulation min avg (inch)	Diameter over braid covering maximum (inch)	Overall diameter minimum (inch)	Overall diameter maximum (inch)	Cold bending mandrel diameter maximum (inches)
-01UN	LSTCJU-4	7/.025	0.105	0.140	0.400	0.430	6
-02UN	LSTCTU-4	7/.025	.105	.140	.400	.430	6
-03UN	LSTCJA-4	7/.025	.105	.140	----	.480	6
-04UN	LSTCTA-4	7/.025	.105	.140	----	.480	6

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Voltage withstand - volts, root mean square, minimum	3000
* Insulation resistance - megohms/1000 feet, minimum	
Conductor to conductor	500
ANSI MC96.1 calibration limits of error:	
(a) types LSTCJA and LSTCJU	± 2.2°C or ± 0.75 percent
(b) types LSTCTA and LSTCTU	± 1°C or ± 0.75 percent
Conductor continuity	No failure
Group A:	
Visual and dimensional	No failure
Watertightness - see MIL-C-24643 for limits of water leakage	No failure
Abrasion resistance - minimum	250
Crack resistance -	No damage
Group B:	
* Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200°C)	50
Gas Flame - 1 hour	No failure
Drip - 95 ± 1°C	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Cold bending, cable - maximum mandrel diameter 12 times maximum cable diameter	No damage
Physicals (unaged)	
Insulation (extruded)	
Tensile strength - lb/in^2, minimum	750
Elongation - percent, minimum	125
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160

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EXAMINATION AND TESTS (continued):

	Requirements
Group C:	
Physicals (aged)	
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged minimum	60
Hot oil immersion	
* Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged minimum	50
Shrinkage	No failure
Heat distortion - percent of unaged, maximum	30
* Permanence of printing (insulation) - cycles, minimum	125
* Permanence of printing (jacket) - cycles, minimum	125
* Cable sealant removability	No failure
Armor (type LSTCTA and LSTCJA only) - conformance to material construction and coverage	No failure
Group D:	
Flame propagation (cable)	No failure

QUALIFICATION INSPECTION:

* Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Cold working (minus 20 ± 2°C)	No damage
Gas Flame (3 hours)	No failure
Accelerated service - sizes 50 through 800 (see table I for load current)	No failure
Aging and compatibility (cable) (125 ± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged minimum	50
Smoke index, maximum	
Jacket	25
* Fillers	45
Insulation	35
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

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UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LSTCJU-4 AND LSTCTU-4	2000

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